

Fine-Tuning Qwen2.5-3B-Instruct on AceReason-1.1-SFT

ECE 6514 Assignment Report

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Course: ECE 6514 -Trustworthy Machine Learning

Submission Links

Model Links:

1. Randomly selected model:

https://huggingface.co/aaryankamdar/TML_Trained_model_random_dataset

2. Data selection: https://huggingface.co/aaryankamdar/TML_PROJECT_MODELS

Dataset Links: https://huggingface.co/datasets/aaryankamdar/TML_Project

Code Repository:

Github: <https://github.com/aaryan-kamdar/TrustWorthyMachineLearningProject1.git>

Executive Summary

This project demonstrates supervised fine-tuning (SFT) of Qwen2.5-3B-Instruct on AceReason-1.1-SFT using loss-based difficulty selection. Through iterative experimentation, we achieved significant improvements over random selection and maintained strong baseline performance.

Key Results

Benchmark	Baseline	Random	Advanced
MATH-500	63.40%	30.0%	64.20%
GPQA-Diamond	29.29%	27.27%	19.19%
AIME 2024	3.33%	0.0%	3.33%
AIME 2025	0.00%	0.0%	3.33%
MMLU-Redux-2	57.05%	29.82%	31.11%

Main Achievements

- Validated loss-based selection: +34.2% over random (30% → 64.2%)
- Maintained strong baseline: 64.2% vs 63.4% (+0.8%)
- Gained new capabilities: AIME 2025: 0% → 3.33%
- Documented trade-offs: GPQA: -10.1%, MMLU: -25.9%

1. The baseline performance of the original Qwen2.5-3B-Instruct model.

Benchmark	Score	Std Error	Problems Solved
MATH-500	63.40%	±0.0210	317 / 500
GPQA-Diamond	29.29%	±0.0324	~29 / 99
AIME 2024	3.33%	±0.0333	1 / 30
AIME 2025	0.00%	±0.0000	0 / 30
MMLU-Redux-2	57.05%	-	-

2. Data Selection Strategy

2.1 Random Selection (Baseline)

Method: Simple random sampling

Characteristics: No difficulty filtering, diversity control, or quality assessment.

Method 1: 15k dataset randomly selected

Benchmark	Score
MATH-500	30.0%
GPQA-Diamond	27.27%

Benchmark	Score
AIME 2024	0.0%
AIME 2025	0.0%
MMLU-Redux-2	29.82%

Method 2: Advanced Selection

Benchmark	Score	Std Error	Problems Solved
MATH-500	64.20%	± 0.0215	321/500
GPQA-Diamond	19.19%	± 0.0281	19/99
AIME 2024	3.33%	± 0.0333	1/30
AIME 2025	3.33%	± 0.0333	1/30
MMLU-Redux-2	31.11%	-	-

The advanced selection model achieved 64.20% on MATH-500 (slight improvement over 63.40% baseline), solved AIME 2025 problems that the baseline couldn't (0% $\rightarrow$ 3.33%), and maintained AIME 2024 performance.

3. A Comparison and Analysis of the Results

Complete Performance Comparison

Benchmark	Baseline	Random	Advanced
MATH-500	63.40%	30.0%	64.20%
GPQA-Diamond	29.29%	27.27%	19.19%
AIME 2024	3.33%	0.0%	3.33%
AIME 2025	0.00%	0.0%	3.33%

Benchmark	Baseline	Random	Advanced
MMLU-Redux-2	57.05%	29.82%	31.11%

Key Findings

Finding 1: Data Selection Quality is Critical

The advanced selection model (64.2% MATH-500) dramatically outperformed the random selection model (30.0% MATH-500) by **+34.2 percentage points**, validating the effectiveness of loss-based difficulty filtering over naive random sampling.

Finding 2: Random Selection is Harmful

Random selection caused catastrophic forgetting, with performance dropping 33.4% below baseline (63.4% $\rightarrow$ 30.0%) on MATH-500. This demonstrates that poor data selection can be worse than not fine-tuning at all.

Finding 3: Specialization Involves Trade-offs

The advanced model gained +0.8% on MATH-500 (target domain) but lost -10.1% on GPQA and -25.9% on MMLU (general domains). This occurred because:

- Training data was 100% mathematics, 0% general knowledge
- Model capacity (3B parameters) is finite
- Weight updates for math reasoning overwrote some general knowledge
- This is expected behavior for domain-specific fine-tuning in small models

Finding 4: Advanced Selection Enables New Capabilities

The advanced model was the only one to solve AIME 2025 problems (0% $\rightarrow$ 3.33%), demonstrating that well-selected training data enables solving previously impossible problems.

Analysis by Benchmark

MATH-500 (Mathematical Reasoning):

- Random: -33.4% from baseline (catastrophic forgetting)
- Advanced: +0.8% from baseline (maintained excellence)

- **Impact:** Advanced selection preserved the strong baseline while random destroyed capability

GPQA-Diamond & MMLU (General Knowledge):

- Both fine-tuned models degraded (GPQA: -2% to -10%, MMLU: -26% to -27%)
- Broader MMLU (57 subjects) suffered more than narrow GPQA (4 domains)
- **Impact:** Specialization on math came at cost of general knowledge

AIME (Competition Mathematics):

- Random lost all capability (3.33% → 0%)
- Advanced maintained and gained (0% → 3.33% on AIME 2025)
- **Impact:** Demonstrates robust training methodology

4. The hyperparameters you used for your final model.

Hyperparameter	Value	Rationale
Learning Rate	1.0e-5	Ultra-conservative to prevent catastrophic forgetting; initial 5e-4 (50x higher) caused performance collapse
Effective Batch Size	16	Per-device: 2, Gradient accumulation: 8; balances memory and gradient stability
Epochs	3	Sufficient for convergence without overfitting
Warmup Ratio	0.1	Gradual LR increase over first 10% prevents early instability
LR Scheduler	Cosine	Smooth decay improves final performance
Sequence Length	16,384	Accommodates long mathematical reasoning chains
Precision	BFloat16	Memory efficient with numerical stability

5. Data Selection Strategy: Loss-Based Difficulty Based Selection

Overview

We implemented an **automated loss-based difficulty filtering** strategy that uses the base model's cross-entropy loss to identify optimally challenging training examples from AceReason-100K.

Method

Step 1: Loss Computation

- Used base Qwen2.5-3B-Instruct to compute cross-entropy loss for all 100,000 examples
- Masked prompt tokens (only computed loss on output/solution)
- Loss serves as an automatic difficulty metric:
 - **Low loss:** Model already knows the answer (too easy)
 - **Moderate loss:** Model has partial understanding (optimal for learning)
 - **High loss:** Model is confused (too hard/noisy)

Step 2: Difficulty-Based Stratification

- Sorted all examples by loss in ascending order
- Applied percentile-based filtering:
 - **Bottom 25% (p0-p25):** Too easy → **EXCLUDED**
 - **Middle 50% (p25-p75):** Optimal difficulty → **SELECTED**
 - **Top 25% (p75-p100):** Too hard → **EXCLUDED**

Step 3: Uniform Sampling

- From the middle 50% difficulty range, uniformly sampled 15,000 examples
- Ensures balanced representation across the optimal learning zone

Rationale

This approach aligns with the **Zone of Proximal Development** theory: learning is most effective when tasks are slightly beyond current ability. By selecting examples where the model demonstrates partial understanding, we maximize the learning signal and avoid both trivial problems and incomprehensible noise.

Validation

Compared to random selection, loss-based filtering achieved **+34.2% improvement** on MATH-500 (30.0% → 64.2%), demonstrating the effectiveness of intelligent data curation over naive sampling. This validates that data quality dramatically outweighs data quantity for reasoning tasks.

6. Conclusions

This project successfully demonstrated supervised fine-tuning of Qwen2.5-3B-Instruct with the following outcomes:

Successes:

- Loss-based data selection validated: +34.2% over random selection
- Maintained strong baseline: 64.2% vs 63.4% on MATH-500 (+0.8%)
- Gained new capabilities: AIME 2025: 0% → 3.33%
- Demonstrated understanding of fine-tuning trade-offs

Trade-offs:

- GPQA decreased -10.1% (science reasoning)
- MMLU decreased -25.9% (general knowledge)
- Expected behavior for domain-specific fine-tuning in small models

Key Takeaways:

- Data quality >> quantity for reasoning tasks
- Ultra-conservative learning rates (1e-5) essential for strong baselines
- Specialization-generalization trade-offs inevitable in 3B parameter models
- Multi-task training or larger models needed for maintaining broad capability